

A Tensor-Product Transfer Subcase For Super Daugavet Points

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Source And Target

Rainis Haller, Johann Langemets, Yoel Perreau, and Triinu Veeorg, *Unconditional bases and Daugavet renormings*, arXiv:2303.07037.

In Section 3, after proving that super Δ -points pass to projective tensor products, the source paper records the following open tensor-transfer issue.

Then

$$T(x_\alpha \otimes y) = \langle Ty, x_\alpha \rangle \rightarrow \langle Ty, x \rangle = T(x \otimes y),$$

because x_α converges to x weakly, and the conclusion follows. $\square$

We do not know whether super Daugavet points pass to the projective tensor product similarly to Daugavet points. However, let us point out that there is an analogue to Lemma 3.12 for those points.

The sentence reads: “We do not know whether super Daugavet points pass to the projective tensor product similarly to Daugavet points.”

Partial Result

Theorem 1. *Let X and Y be Banach spaces over $\mathbb{K}$. Suppose $x_0 \in S_X$ is a super Daugavet point. Suppose $y_0 \in S_Y$ spans a one-dimensional L -summand of Y , i.e. there is a Banach space Z such that*

$$Y = \mathbb{K}y_0 \oplus_1 Z$$

isometrically. Then $x_0 \otimes y_0$ is a super Daugavet point of $X \widehat{\otimes}_\pi Y$.

Corollary 1. *If x_0 is a super Daugavet point in X , then:*

- for every set Γ and every $\gamma \in \Gamma$, $x_0 \otimes e_\gamma$ is a super Daugavet point of $X \widehat{\otimes}_\pi \ell_1(\Gamma)$;*
- if A is an atom of finite positive measure in a measure space (Ω, Σ, μ) , then $x_0 \otimes \mu(A)^{-1} \mathbf{1}_A$ is a super Daugavet point of $X \widehat{\otimes}_\pi L_1(\mu)$.*

Idea Of The Proof

The key point is that the projective tensor product distributes isometrically over an ℓ_1 -sum. If $Y = \mathbb{K}y_0 \oplus_1 Z$, then

$$X \widehat{\otimes}_\pi Y \cong X \oplus_1 (X \widehat{\otimes}_\pi Z),$$

and $x_0 \otimes y_0$ becomes $(x_0, 0)$. Thus the problem reduces to a simple stability fact: a super Daugavet point remains super Daugavet after embedding it as the first coordinate of an ℓ_1 -sum.

This only treats tensor factors with a one-dimensional ℓ_1 -summand. It does not settle the source paper's general question for arbitrary $y_0 \in S_Y$.

Proof

We use the following standard net characterization, included for completeness.

Lemma 1. *Let $x \in S_X$. Then x is a super Daugavet point if and only if, for every $u \in B_X$, there is a net $(u_\alpha) \subset B_X$ such that*

$$u_\alpha \xrightarrow{w} u \quad \text{and} \quad \|x - u_\alpha\| \longrightarrow 2.$$

Proof. If x is a super Daugavet point, order pairs (U, ε) , where U is a weak neighbourhood of u in B_X and $\varepsilon > 0$, by reverse inclusion and decreasing ε . For each pair choose $u_{U, \varepsilon} \in U$ with $\|x - u_{U, \varepsilon}\| > 2 - \varepsilon$. This gives the desired net.

Conversely, let W be a non-empty relatively weakly open subset of B_X and choose $u \in W$. A net as above is eventually in W , so $\sup_{w \in W} \|x - w\| = 2$. $\square$

Lemma 2. *Let x_0 be a super Daugavet point in X , and let Z be any Banach space. Then $(x_0, 0)$ is a super Daugavet point in $X \oplus_1 Z$.*

Proof. Take $(u, v) \in B_{X \oplus_1 Z}$, and put $s = \|v\|$ and $r = 1 - s$. Then $\|u\| \leq r$.

If $r = 0$, then $u = 0$, and the constant net $(0, v)$ already satisfies

$$\|(x_0, 0) - (0, v)\| = \|x_0\| + \|v\| = 2.$$

Assume now that $r > 0$. By Lemma 1, choose a net $(p_\alpha) \subset B_X$ such that

$$p_\alpha \xrightarrow{w} u/r \quad \text{and} \quad \|x_0 - p_\alpha\| \rightarrow 2.$$

Set $w_\alpha = (rp_\alpha, v)$. Then

$$\|rp_\alpha\| + \|v\| \leq r + s = 1,$$

so $w_\alpha \in B_{X \oplus_1 Z}$, and clearly $w_\alpha \xrightarrow{w} (u, v)$. Moreover,

$$\begin{aligned} \|(x_0, 0) - w_\alpha\| &= \|x_0 - rp_\alpha\| + s \\ &\geq \|x_0 - p_\alpha\| - (1 - r)\|p_\alpha\| + s \\ &\geq \|x_0 - p_\alpha\|. \end{aligned}$$

The reverse inequality $\|(x_0, 0) - w_\alpha\| \leq 2$ is automatic because both points lie in the unit ball. Hence the distances tend to 2, and Lemma 1 proves that $(x_0, 0)$ is super Daugavet. $\square$

We now prove Theorem 1. The projective tensor product is isometrically additive over ℓ_1 -sums:

$$X \widehat{\otimes}_\pi (\mathbb{K}y_0 \oplus_1 Z) \cong (X \widehat{\otimes}_\pi \mathbb{K}y_0) \oplus_1 (X \widehat{\otimes}_\pi Z) \cong X \oplus_1 (X \widehat{\otimes}_\pi Z).$$

Under this canonical isometry, $x_0 \otimes y_0$ corresponds to $(x_0, 0)$. The conclusion follows directly from Lemma 2. $\square$

The corollary follows because each coordinate line in $\ell_1(\Gamma)$ is an L-summand, and a finite positive atom A gives the isometric decomposition

$$L_1(\mu) = \mathbb{K}\mu(A)^{-1}\mathbf{1}_A \oplus_1 L_1(\mu|_{\Omega \setminus A}).$$

Scope And Limitations

This is a partial positive answer to the tensor-transfer uncertainty in the source paper. It covers tensor factors with an L-summand coordinate, including $\ell_1(\Gamma)$ coordinates and atoms of $L_1(\mu)$. It does not prove that $x_0 \otimes y_0$ is super Daugavet for an arbitrary Banach space Y and arbitrary $y_0 \in S_Y$.

This packet also does not address the later renorming question in the source paper asking whether ℓ_1 can be renormed to contain a super Daugavet point.

Novelty And Duplicate Check

Local duplicate search was run over `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` for arXiv:2303.07037 and the phrases “super Daugavet”, “projective tensor”, and “Daugavet renormings”. No existing packet for this source paper or this tensor subcase was found.

External searches were bounded to phrases including “super Daugavet projective tensor”, “super-Daugavet projective tensor product”, and “super Daugavet points pass projective tensor”. No exact later answer to this subcase was found. Novelty confidence is therefore modest: the theorem is elementary once the L-summand reduction is isolated, but it records a concrete positive class for the source paper’s open tensor-transfer question.

References

References

- [1] R. Haller, J. Langemets, Y. Perreau, and T. Veeorg, *Unconditional bases and Daugavet renormings*, arXiv:2303.07037.
- [2] R. A. Ryan, *Introduction to Tensor Products of Banach Spaces*, Springer Monographs in Mathematics, Springer, 2002.